Dense, reliable, and energy efficient interconnects for AI

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What does the AI industry need?

AI is rapidly driving innovation to meet the industry's needs

There is a lot of activity and innovation in the industry to develop the needed solutions

When standardization is desired, there are two general paths:

- MSAs (Multi-Source Agreements)
  - MSAs are faster to publish, have restricted membership, but are typically less well vetted
- SDOs (Standards Development Organizations)
  - SDOs develop very detailed interoperability agreements, but are generally slower

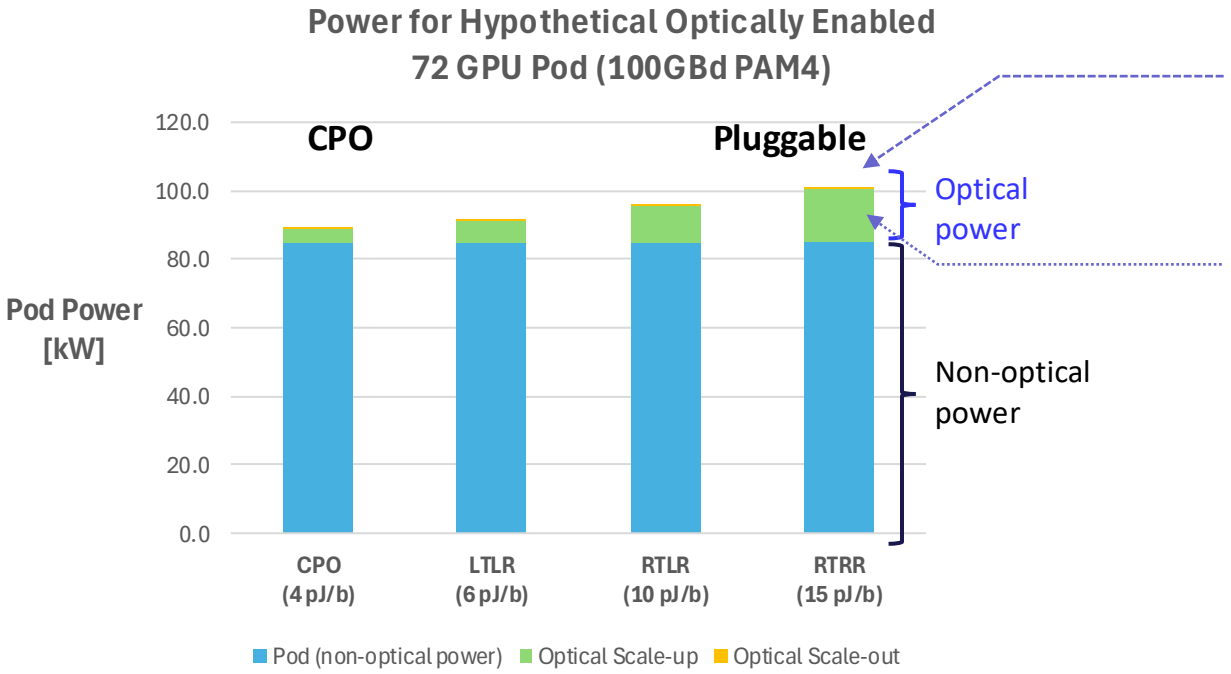
What is needed?

- Standards developed on "AI timelines"
- That meet the application requirements
- That enable secure supply

Standards developed on AI Timelines

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Scale-up benefits from interconnect energy efficiency



Pod Configuration	Scale-out	Scale-up
Number of Optical Links	72	2592
Link capacity [G]	800	400
Total bandwidth [T]	57.6	1036.8

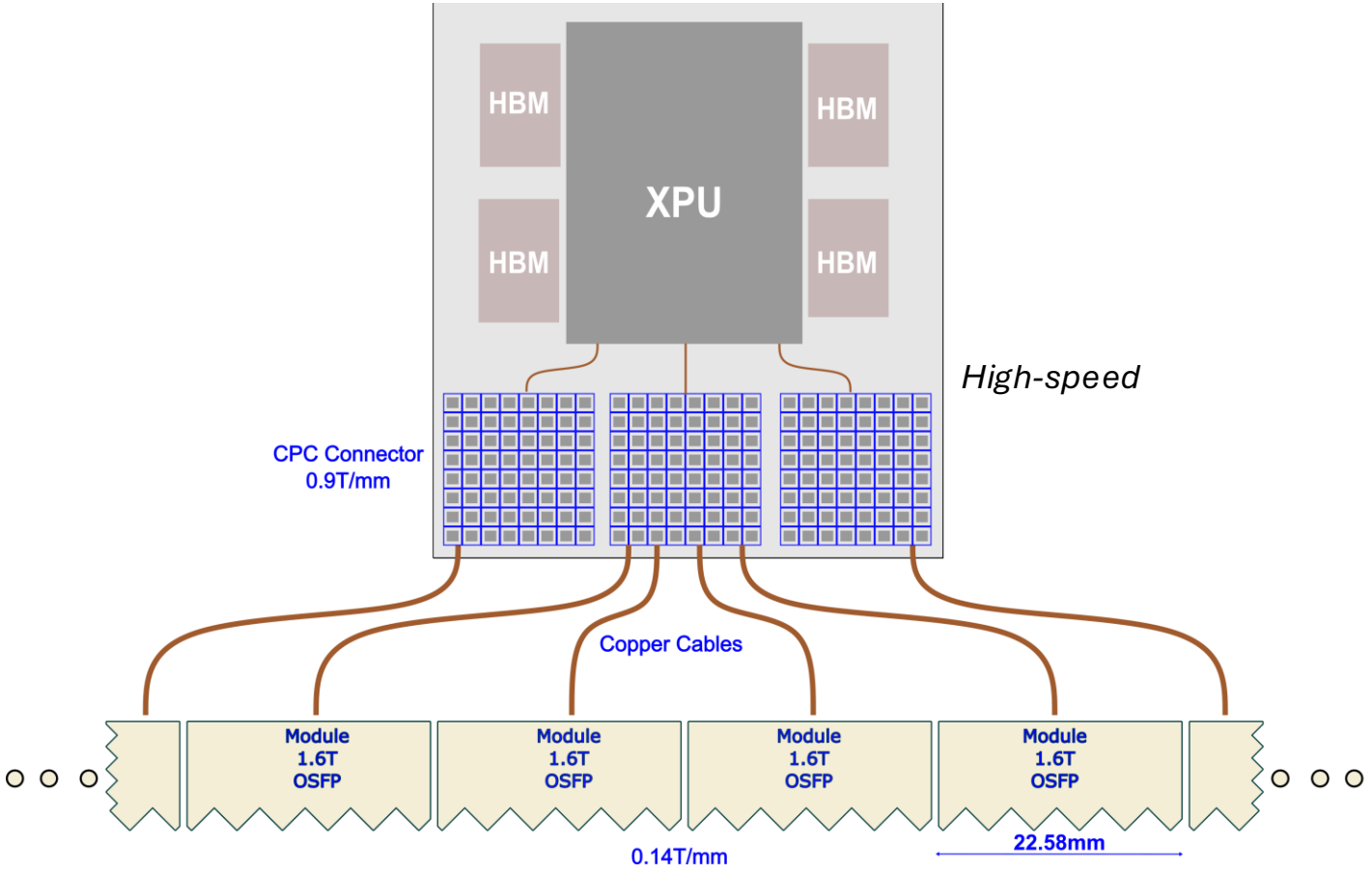
- Scale-out networking bandwidth is about ~1/10x that of scale-up networks and fewer links
- Scale-up dominates the rack interconnect power consumption
- The lowest power solutions for optical connectivity will leverage co-packaging or lower power options for scale-up

Scale-up dominates the rack interconnect power
The lowest power solutions for optical connectivity will leverage CPO or lower power options for scale-up

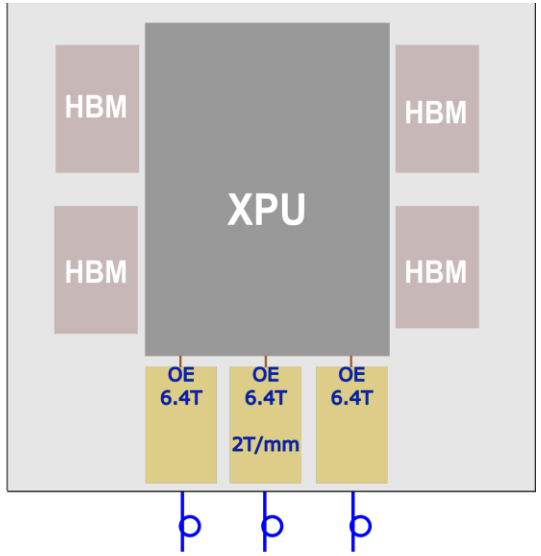
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Low power drives shorter electrical channels

OSFP Pluggables
for 19.2T



CPO
19.2T



Note: $BW_{DENSITY} = \frac{BW_{Tx+Rx}}{W_{EDGE}}$

High-speed or D2D I/Fs

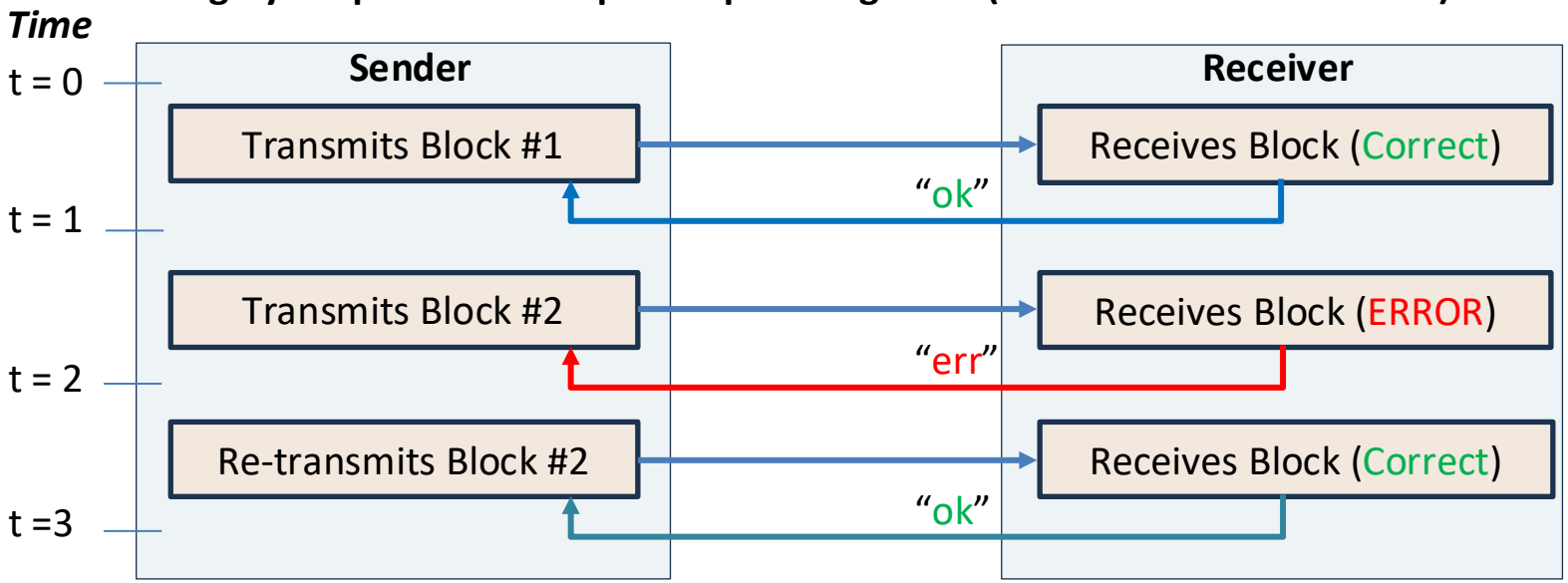
New Denser Options

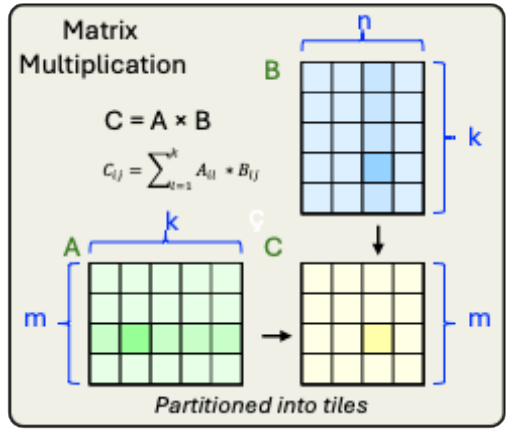
The industry is advancing denser options such as the front-pluggable XPO (XPO-MSA) form factor and the OIF's High-Density 12.8T NPO project

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Why reducing link errors for scale-up is important

Highly simplified scale-up example using CBFC (credit-based flow control)





Job partitioned into tiles assigned to accelerators (exchanging large blocks of data)

- The following block is only transmitted once the current block is received correctly
- Periodic retransmissions drive tail latency & reduces effective interconnect BW
- Link flaps occur when the number of retries exceeds the retry limit → focus on minimizing burst errors

Minimizing burst errors to prevent link flaps is a key objective to reduce tail latency for scale-up

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What makes a compelling solution for AI Scale-up?

Radix

It's not about huge pipes, but achieving target radix

$$Radix = \frac{Capacity}{BW\ per\ pipe}$$

Shoreline Density

Dense enough for electrical signals to escape the ASIC

$$BW_{DENSITY} = \frac{BW_{Tx+Rx}}{W_{EDGE}}$$

Latency Tail latency

Minimize latency and variation in latency

Link Reliability

Minimize link errors/flaps

Hardware Reliability

Minimize hardware failures through leveraging standard silicon eco-system for high reliability

Energy Utilization

Minimize energy usage
Use low power options with short channels

High Volume MFG

Support steep production ramps with excellent C_{PK}
Need predictable supply chain

Total Cost of Ownership

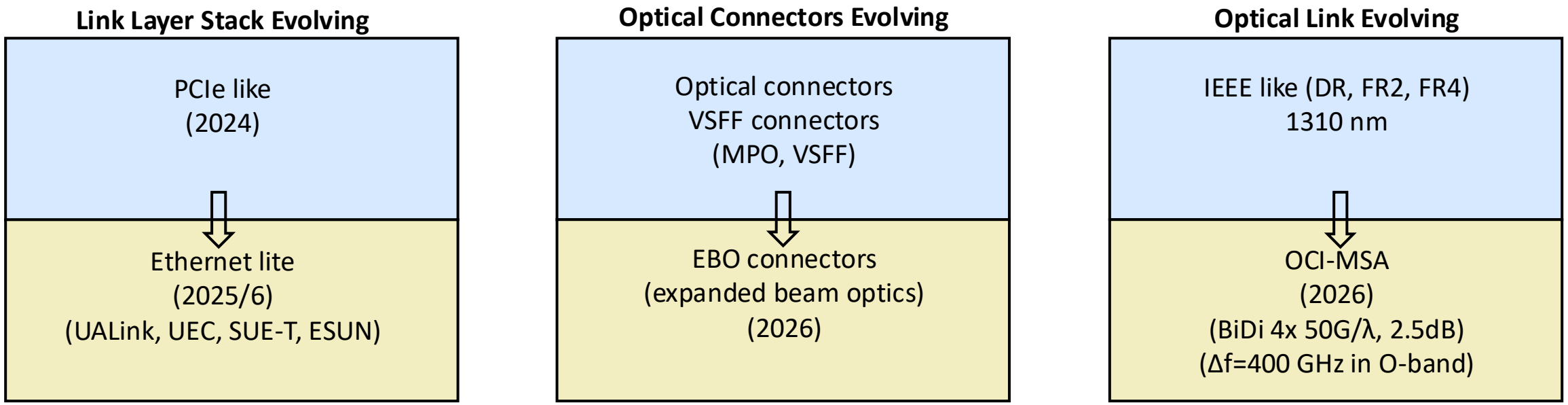
Will evolve to the lowest cost implementation
(acquisition, install, downtime, retries, repair, power consumption, etc.)

Scalability

Technology scales to next generation

There are many good ideas
The ones that are adopted will meet the application's criteria

Reality: AI scale-up has been a bit of a moving target

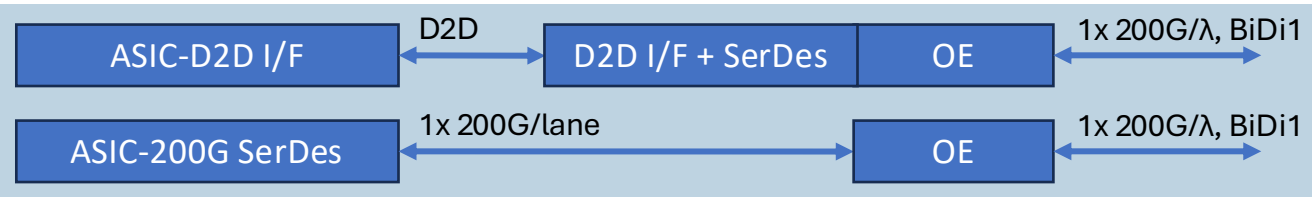


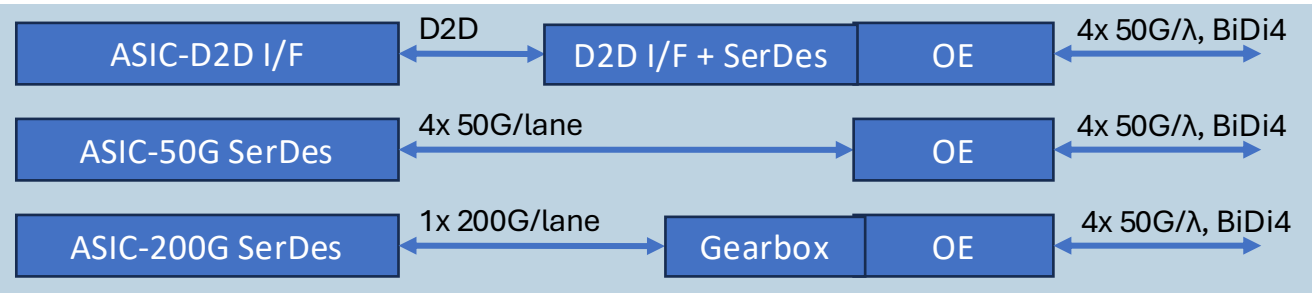
One major challenge for standards is maintaining alignment with this fast-moving space

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Some possible 200G/F implementations for scale-up



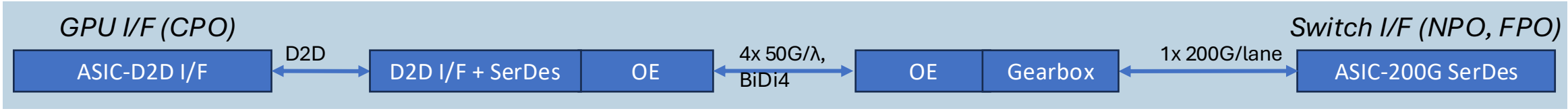




Type	Electrical	Optical	FormFactor	Limitations
PSM	1x 200G/L	1x 200G/λ PSM	CPO, NPO, pluggable	• Fiber count (2x)
COI-BiDi1 ‡	D2D	1x 200G/λ BiDi1 Δλ = 20nm	CPO	• CPO Only
	1x 200G/L		CPO, NPO, pluggable	
OCI G1 (BiDi4)	D2D	4x 50G/λ BiDi4 Δf = 400GHz	CPO	• CPO Only
	4x 50G/L		CPO, NPO, pluggable	• Shoreline density
	1x 200G/L		CPO, NPO, pluggable	• Energy utilization

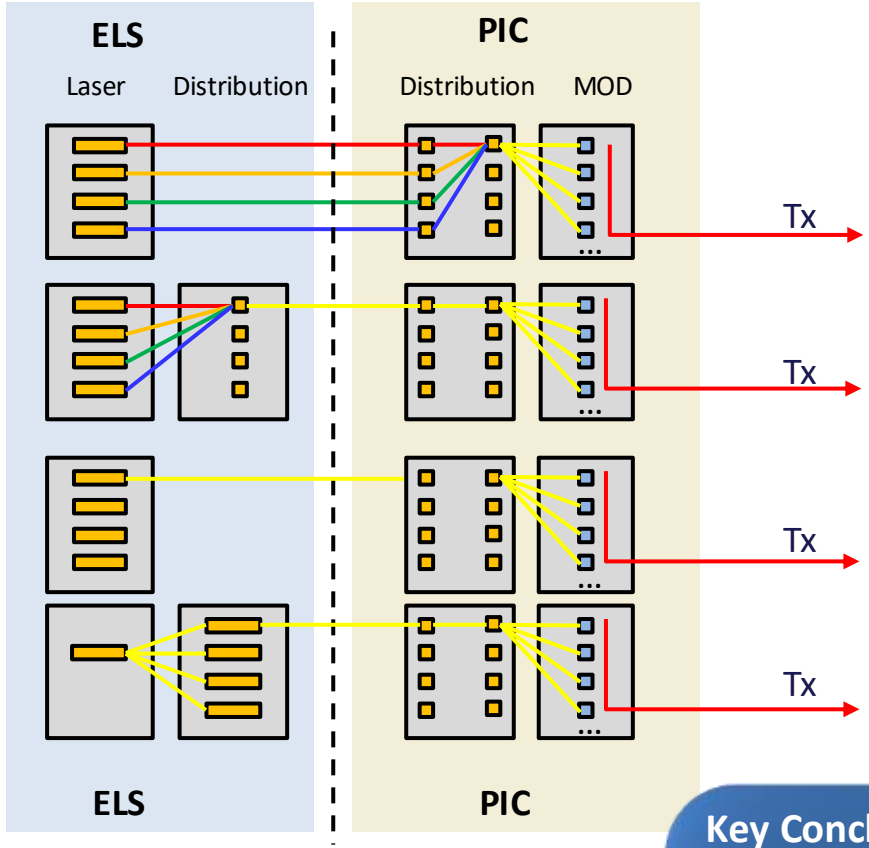
‡: COI-BiDi1 is speculative

Example of a mixed CPO-NPO link for OCI G1



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External WDM laser/OE architectural considerations



Cost differences

- Adding extra functionality in the PIC should be more cost effective than a standalone mux in the ELS as shoreline density requirements are met

Reliability differences

- The PIC with additional functionality should be much more reliable than placing that functionality in the ELS

Blast radius differences

- Shared functions across laser outputs in a multi-OE ELS increase blast radius
- An ELS which support >1 OE has a higher blast radius unless that amount of "shared" components is significantly minimized

Scalability to high volume

- Some ELS technologies still need to be proven they can scale to high-volume

Non-linear effects

- Optical connectors and splitters carrying high-power multi-λs can experience non-linear effects

Notes:

1. Assumes the optical power at the modulator is the same for the different configurations
2. Optical connectors and waveguides have limitations for the total power per fiber or waveguide

Key Conclusions:

- **Placing mux and splitting functions into the PIC improves reliability, manufacturability, and cost**
- **Deploy the ELS in a 1:1 relationship to minimize blast radius**
- **Until the industry matures, set Tx & Rx specifications to enable multiple technologies and to stimulate innovation**

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Summary

AI is driving rapid innovation especially for scale-up optics

- Standards efforts need to adjust to the new reality and differing requirements

There are many criteria that a scale-up solution needs to meet

- It's not just about a unique technology, but one that not only meets the required performance but also can scale rapidly, has low cost, and offers excellent reliability

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